

# Technical Document #596: Natural Language Processing - Comprehensive Overview

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Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

### Key Takeaways:

- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.
- Coreference resolution identifies all expressions that refer to the same entity in a text.
- Word embeddings represent words as dense vectors in continuous space.